VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 13 | JEE - 2022

Date: 29/09/2021 **Maximum Marks: 300**

Timing: 04:00 PM to 09:00 PM

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

Candidate's Signature : _ _ _ _ _ _ _ _ _ _ **Invigilator's Signature** _ _ _ _ _ _ _ _

PART I : PHYSICS — 100 MARKS

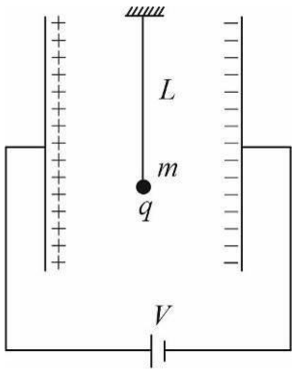
SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A $5\,\mu F$ capacitor is fully charged to a potential difference of 20 V. After removing the source voltage, it is connected to an uncharged capacitor of $10\,\mu F$ in parallel. The charge on the second capacitor is :

(A) $50\,\mu C$ **(B)** $100/3\,\mu C$ **(C)** $200/3\,\mu C$ **(D)** $40\,\mu C$

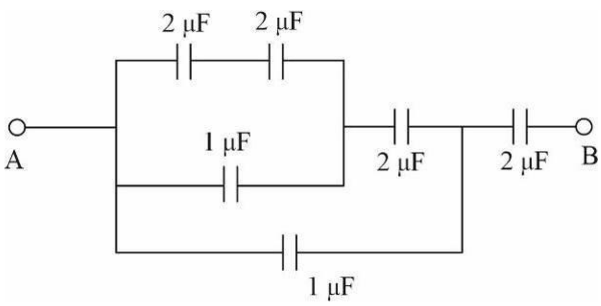
2. A simple pendulum of length L is placed between the plates of a parallel plate capacitor having area of plates A, distance between the plates d and connected to a battery of potential difference V. If the bob has a mass m and charge q, find time period of simple pendulum.



(A) $2\pi\sqrt{\dfrac{L}{\sqrt{g^2+\left(\dfrac{qV}{md}\right)^2}}}$ **(B)** $2\pi\sqrt{\dfrac{L}{\sqrt{g^2+\left(\dfrac{qV}{mA}\right)^2}}}$

(C) $2\pi\sqrt{\dfrac{L}{g}}$ **(D)** None of these

3. In the diagram shown, the net capacitance between the points A and B is (in μF) :

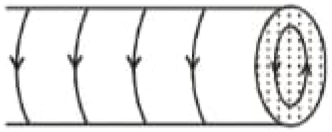


(A) $\dfrac{2}{3}$ **(B)** 1 **(C)** 3 **(D)** $\dfrac{10}{3}$

4. Capacitance of a parallel plate capacitor becomes 5/4 times its original value, if a dielectric slab of thickness $t=\dfrac{d}{3}$ is inserted between the plate [d is the separation between the plates]. The dielectric constant of the slab is :

(A) 4 **(B)** 8 **(C)** 2 **(D)** 2.5

5. Magnetic field is uniform and has a magnitude B in the interior of a very long solenoid far from its ends. One of the ends of the solenoid is closed with a thin flat plastic cover. A single small electrical loop of radius R lies on the cover so that its center is on the axis of the solenoid. The electrical current flowing in the loop in I. Then the mechanical tension in the loop's wire is:



(A) $\frac{BIR}{2}$ **(B)** BIR **(C)** $2\pi BIR$ **(D)** πBIR

6. A conducting body 1 has some initial charge $243\mu C$, and its capacitance is C. There are two other conducting bodies, 2 and 3, having capacitances: $C_2 = 2C$ and $C_3 \to \infty$. Bodies 2 and 3 are initially uncharged. "Body 2 is touched with body 1. Then, body 2 is removed from body 1 and touched with body 3, and then removed." This process is repeated N times.

Choose the incorrect statement(s):

(A) Charge on the body 1 after this process is repeated for the fourth time is $1\mu C$

(B) Charge on the body 1 after this process is repeated for the third time is $9\,\mu C$

(C) The charge removed from body 1 in the fourth time is $6\mu C$

(D) The total charge on body 3 is equal to the total charge removed from body 1

7. Two identical wires A and B, each of length l, carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side `a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is :

(A) $\frac{\pi^2}{16\sqrt{2}}$ **(B)** $\frac{\pi^2}{16}$ **(C)** $\frac{\pi^2}{8\sqrt{2}}$ **(D)** $\frac{\pi^2}{8}$

8. An insulating thin rod of length l has linear charge density $\rho(x) = \rho_0 \frac{x^2}{l}$ on it. The rod is rotated about an axis passing through the origin $(x = 0)$ and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is :

(A) $n\rho_0 l^3$ **(B)** $\frac{\pi}{5} n\rho_0 l^4$ **(C)** $\frac{\pi}{4} n\rho_0 l^4$ **(D)** $\pi n\rho_0 l^3$

9. Two long conductors, separated by a distance d carry currents I_1 and I_2 in the same direction. They exert a force F on each other. Now, the current in one of them is increased to four times and its direction is reversed. The distance is also increased to $4d$. The new value of the force between them is :

(A) $-F$ **(B)** $\frac{F}{3}$ **(C)** $-\frac{2F}{3}$ **(D)** $-\frac{F}{3}$

10. A circular loop of area $10\,m^2$ carrying a current of $10A$, is held with its plane perpendicular to a magnetic field induction $2T$. The torque acting on the loop is :

(A) 200 N-m **(B)** 100 N-m **(C)** zero **(D)** 20 N-m

11. In a certain region static electric and magnetic fields exist. The magnetic field is given by $\vec{B} = B_0(\hat{i} + \hat{j})$. If a test charge moving with a velocity $\vec{V} = V_0(2\hat{i} - \hat{j} + \hat{k})$ experiences no force in that region, then the electric field in the region, in SI units, is :

(A) $V_0 B_0(\hat{i} - \hat{j} - 3\hat{k})$ **(B)** $V_0 B_0(\hat{i} + \hat{j} + 3\hat{k})$

(C) $V_0 B_0(-\hat{i} + \hat{j} + 3\hat{k})$ **(D)** $V_0 B_0(-\hat{i} - \hat{j} - 3\hat{k})$

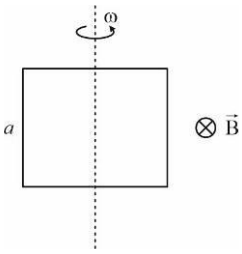
12. The self-induced emf of a coil is 20 volts. When the current in it is changed at uniform rate from 10A to 20A in 1s, the change in the energy of the inductance is:

(A) 637.J **(B)** 100 J **(C)** 400 J **(D)** 300 J

13. An ideal capacitor of capacitance $0.1\mu F$ is charged to a potential difference of 20V. The charging battery is then disconnected. The capacitor is then connected to an ideal inductor of self inductance $1mH$. The current at a time when the potential difference across the capacitor is 10 V, is:

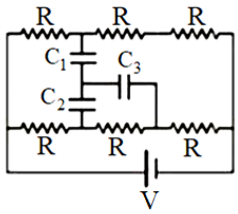
(A) 3A **(B)** 0.17 A **(C)** 1.7 A **(D)** 0.3 A

14. In a uniform magnetic field of induction B, a wire in the form of square of side a rotates about the axis as shown with angular frequency ω. If the total resistance of the circuit is R, the mean power generated per period of rotation is:



(A) $\frac{B^2\omega^2a^2}{2R}$ **(B)** $\frac{B^2\omega^2a^2}{4R}$ **(C)** $\frac{B^2\omega^2a^4}{2R}$ **(D)** $\frac{B\omega a^2}{2R}$

15. In the shown circuit, all three capacitor are identical and have capacitance $C\ \mu F$ each. Each resistor has resistance of $R\ \Omega$. An ideal cell of emf V volts is connected as shown. If the magnitude of potential difference across capacitor C_3 in steady state is:



(A) $\frac{V}{9}$ **(B)** $\frac{2V}{9}$ **(C)** $\frac{V}{3}$ **(D)** $\frac{V}{4}$

16. Two coils have a mutual inductance of 0.001H. The current changes in the first coil according to equation $I = I_0 \sin \omega t$, where $I_0 = 1A$ and $\omega = 10\pi rad/s$. The maximum value of emf (in volt) in the second is :

(A) π **(B)** 10π **(C)** $\frac{\pi}{100}$ **(D)** 2π

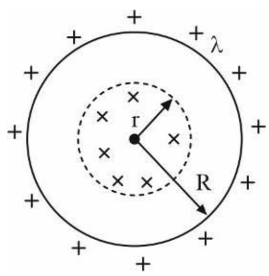
17. The figure shows a non-conducting ring of radius R carrying a uniform linear charge density λ. In a circular region of radius r, a uniform magnetic field $\vec{B}$ perpendicular to the plane of the ring varies at a constant rate $\frac{dB}{dt} = \alpha$. The torque acting on the ring is :

(A) $\alpha\pi r^2\lambda R$

(B) $\alpha\lambda\pi r^2/2$

(C) $\frac{\alpha\lambda\pi R^2 r}{2}$

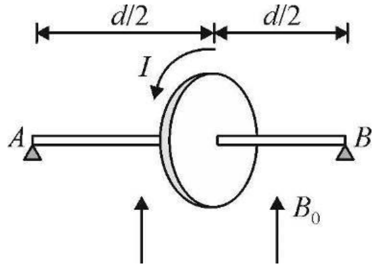
(D) $\frac{\alpha\lambda\pi r^2}{2}R$



18. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by:

(A) $\sqrt{\frac{L}{C}} \times \frac{1}{R}$ **(B)** $\frac{\omega_0 R}{L}$ **(C)** $\frac{R}{(\omega_0 C)}$ **(D)** $\frac{CR}{\omega_0}$

19. A wooden disc of mass M and radius R has a single loop of wire wound on its circumference. It is mounted on a massless rod of length *d*. The ends of the rod are supported at its ends so that the rod is horizontal and disc is vertical. A uniform magnetic field B_0 exists in vertically upward direction. When a current *I* is given to the wire one end of the rod leaves the support. Least value of *I* is :



(A) $2MgR / \pi d^2 B_0$ **(B)** $MgR / 2\pi d^2 B_0$

(C) $2Mgd / \pi R^2 B_0$ **(D)** $Mgd / 2\pi R^2 B_0$

20. An alternating voltage $E = 20 \sin 200t$ is applied to a series combination $R = 60\Omega$ and an inductor of 400 mH. The power factor of the circuit is:

(A) 0.01 **(B)** 0.2 **(C)** 0.05 **(D)** 0.6

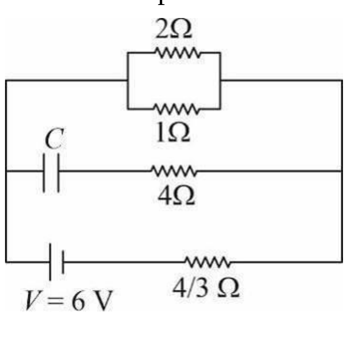
SPACE FOR ROUGH WORK

SECTION-2

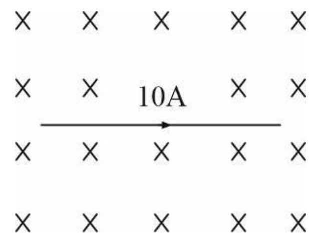
This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. A thin conducting wire in the shape of a 'figure of eight' is situated with its circular loops in two planes making an angle of 120° with each other if the current in the loop is I and the radius is R, the magnetic induction at a point of intersection P of the two axes of the loops is $\frac{N\mu_0 i}{48R}$. Find N.

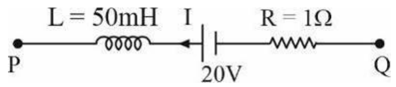
22. Calculate the steady state current (in A) in the 2Ω resistor shown in the circuit (see figure). The internal resistance of the battery is negligible and the capacitance of the condenser C is $0.1\,\mu F$.



23. A current element of length 50 cm is carrying a current of $10\ A$. The region contains a magnetic field of unknown magnitude. The mass of wire is $40\ \text{gm}$. Under the action of gravitational and magnetic forces the wire stays in equilibrium. The value of magnetic field is $x \times 10^{-2} T$, then x will be _______.
(Take $g = 10\ \text{ms}^{-2}$)



24. A part of a complete circuit is shown in the figure. At some instant, the value of current I is 2 A and it is decreasing at a rate of $2 \times 10^2\ As^{-1}$. The value of the potential difference $V_P - V_Q$. (in volts) at that instant is_________.



25. A step down transformer operates at 200 V and current through secondary coil is 15A. If it has turn ratio of 1.5 then current in primary coil (in Amperes) is:

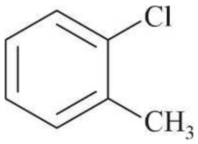
SPACE FOR ROUGH WORK

PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which of the following is correct IUPAC name of the following compound ?



(A) 1-chloro-2-methyl benzene **(B)** 2-chlorotoluene

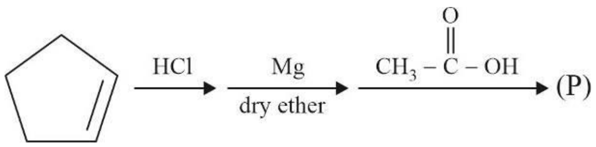
(C) 1-chloro-6-methyl benzene **(D)** Both (A) and (B)

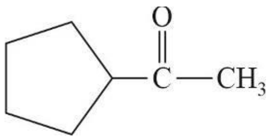
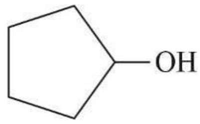
2. Which of the following is correct reactivity order for S_N2 reaction of the following compounds ?

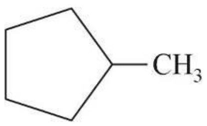
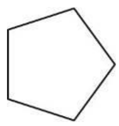
$$\underset{\textbf{I}}{C_6H_5-CH_2Br}, \quad \underset{\textbf{II}}{C_6H_5-\underset{\large Br}{\underset{|}{C}H}-CH_3}, \quad \underset{\textbf{III}}{C_6H_5-\underset{\large Br}{\underset{|}{C}H}-C_6H_5}$$

(A) I < II < III **(B)** III < I < II **(C)** III < II < I **(D)** II < III < I

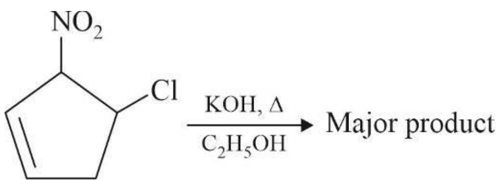
3. Identify major product (P) of the following reaction sequence.

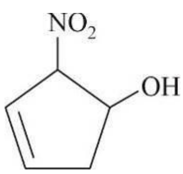
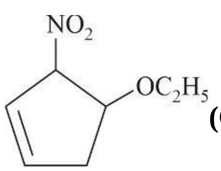
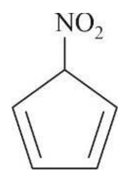
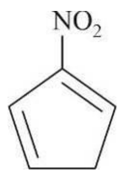


(A)  **(B)** 

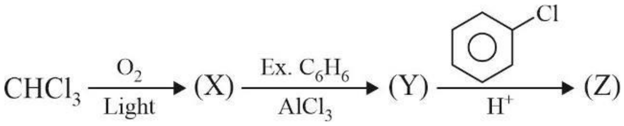
(C)  **(D)** 

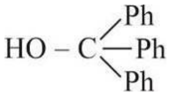
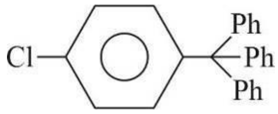
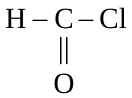
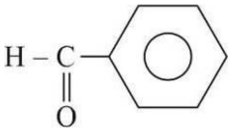
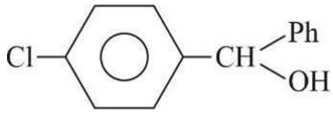
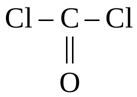
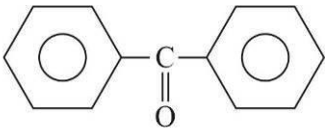
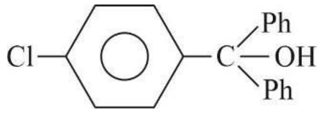
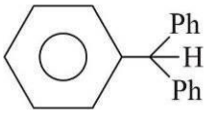
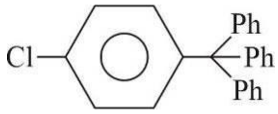
4. The major product of the following reaction is :



(A)  **(B)**  **(C)**  **(D)** 

5. The structure of major product X, Y and Z formed in the following reaction are : (Ph is phenyl group)



	X	Y	Z
(A)	$HO-CCl_3$		
(B)			
(C)			
(D)	CCl_2		

6. Which of the following is not correct ?

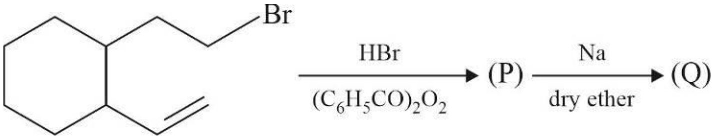
(A) Methylene chloride : used as metal cleaning
(B) Chloroform : used as refrigerant R-22
(C) Iodoform : used as an antiseptic
(D) Carbon tetrachloride : used as fire extinguisher

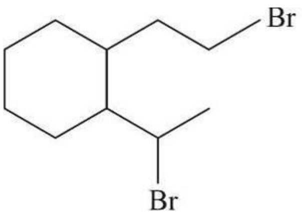
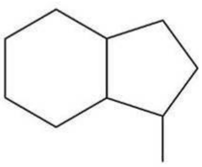
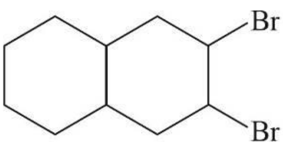
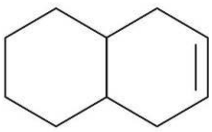
7. Which of the following is not correct for organic product of the following reaction :

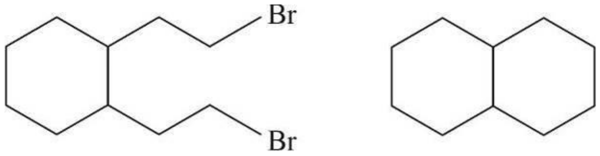
$$\underset{(1\text{mole})}{CCl_4} + \underset{(1\text{mole})}{CoF_2} \longrightarrow \text{Product}$$

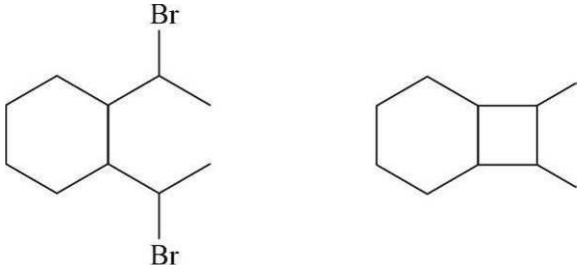
(A) Organic product is extremely stable
(B) Organic product is unreactive and non-toxic
(C) Organic product is non corrosive
(D) Organic product can't be easily liquefied

8. Identify major products (P) and (Q) of the following reaction sequence.

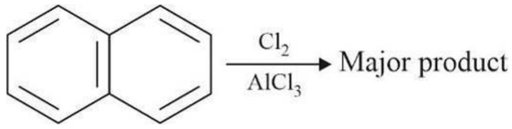


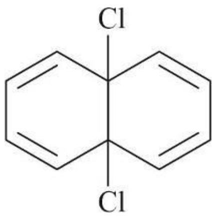
	(P)	(Q)
(A)		
(B)		

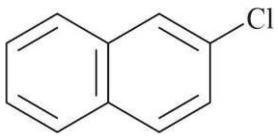
(C) 

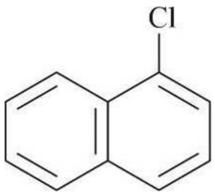
(D) 

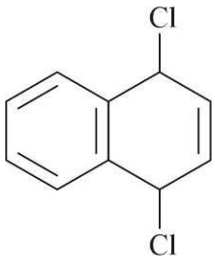
9. Identify major product of the following reaction.



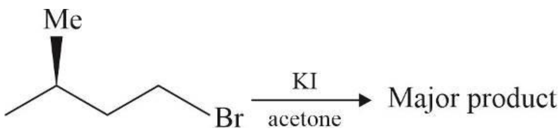
(A) 

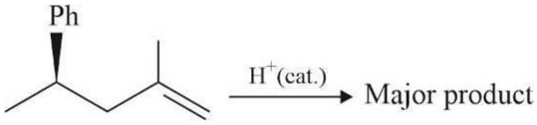
(B) 

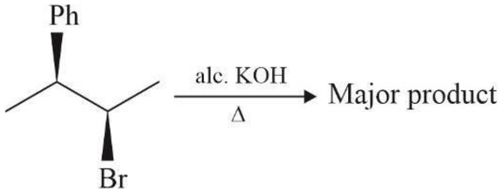
(C) 

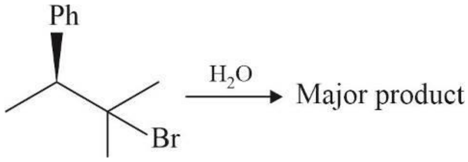
(D) 

10. Which of the following reaction proceed with retention in configuration of chiral carbon atom ?

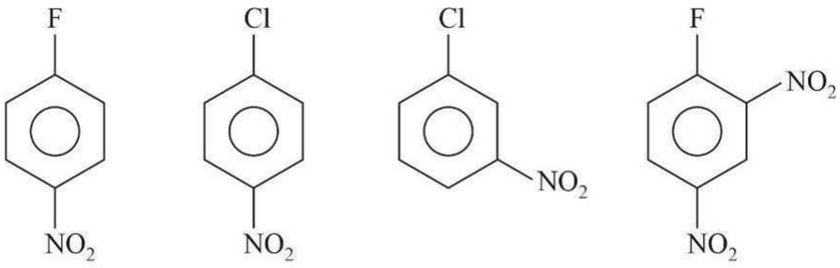
(A) 

(B) 

(C) 

(D) 

11. Which of the following is correct reactivity order of the following aryl halides with hot and conc. NaOH ?

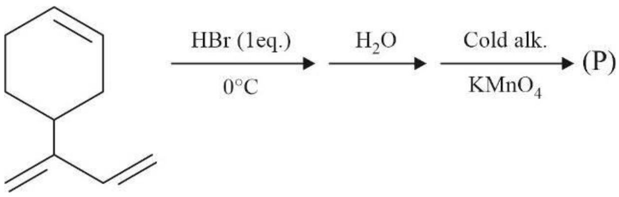


I **II** **III** **IV**

(A) III > II > IV > I **(B)** IV > I > II > III

(C) II > III > I > IV **(D)** I > II > III > IV

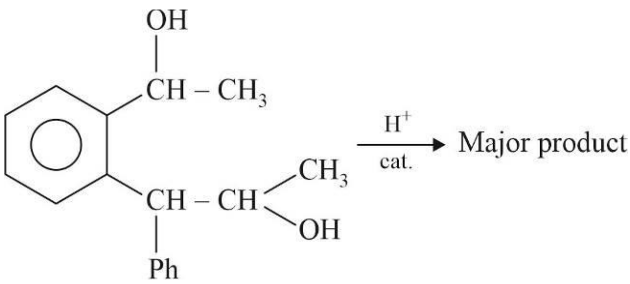
12.



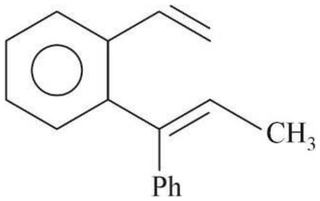
Find the number of OH group in product (P).

(A) 6 **(B)** 4 **(C)** 5 **(D)** 3

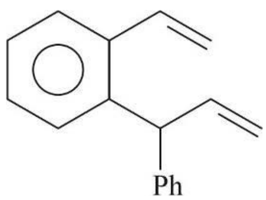
13. Identify major product of the following reaction.



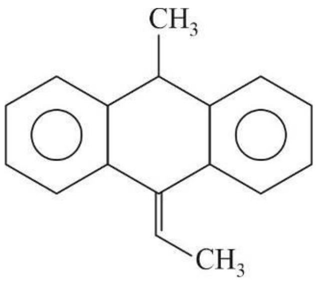
(A)



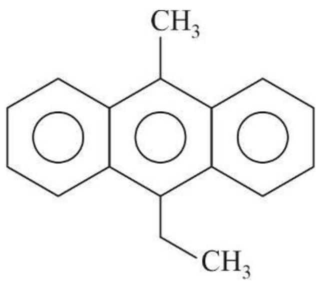
(B)



(C)



(D)



14. A hydrocarbon does not react with chlorine (Cl_2) in dark but gives a single monochloro compound in bright sunlight. Identify the hydrocarbon.

(A) Ethene **(B)** Cyclopentane **(C)** 2-Butene **(D)** Pentane

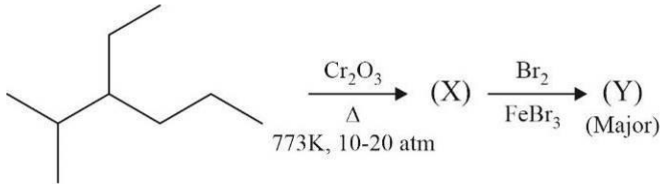
15. Which of the following is not correct order ?

(A) $CH_3-Br > CH_3-F$; Reactivity for S_N2

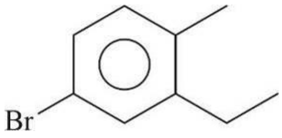
(B) $CH_3-Br > CH_3-F$; Order of dipole moment

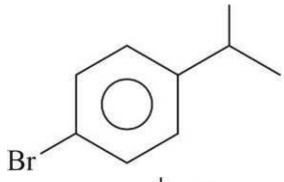
(C) $CH_3-Br > CH_3-F$; Order of boiling point

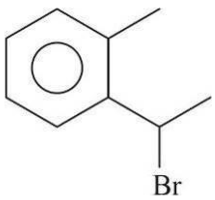
(D) $CH_3-Br > CH_3-F$; Order of density

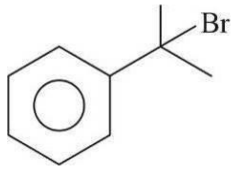
16. 

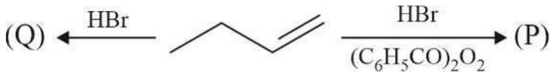
(X) on oxidation by acidified hot $KMnO_4$ produce monocarboxylic acid. Identify major product (Y).

(A) 

(B) 

(C) 

(D) 

17. 

Which of the following is not correct for major organic bromides (P) and (Q) ?

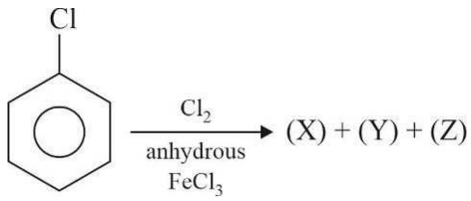
(A) (P) and (Q) are positional isomers

(B) Boiling point of (Q) is 364K and boiling point of (P) is 375K

(C) (P) and (Q) form same product on heating with potassium tert-butoxide

(D) (P) and (Q) both are chiral

18. Which of the following is not correct for disubstituted products (X), (Y) and (Z), if order of percent yield is (X) > (Y) > (Z) ?



(A) Order of boiling point : (Y) > (X) > (Z)

(B) Order of melting point : (X) > (Y) > (Z)

(C) Order of dipole moment : (Y) > (Z) > (X)

(D) Order of reactivity with NaOH(aq) at 25°C and 1 atm : (Z) > (X) > (Y)

19. Which of the following is correct for S_N2 reaction of chiral substrate ?

(A) It is regioselective

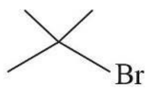
(B) It is regiospecific

(C) It is stereo selective as well as stereospecific

(D) It is neither stereoselective nor stereospecific

20. Which of the following is not correct for isomeric bromo butanes. I and II.

I



II

(A) (I) is most suitable for S_N2

(B) (II) is most suitable for S_N1

(C) (II) is more reactive than (I) for E2

(D) (I) is more reactive than (II) for E1

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. $\underset{\substack{C_4H_9Br \\ \text{(Primary bromide)}}}{(A)} \xrightarrow[\text{dry ether}]{Na} (D)$

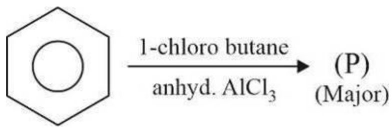
(D) can't be prepared by reaction of n-butyl bromide with sodium in dry ether.
Find the number of primary carbon atoms in (D).

22. An ester $(A)(C_4H_8O_2)$ on reaction with excess of CH_3MgBr in dry ether followed by acidification produce $(B)(C_3H_8O)$ as sole organic product. (B) form yellow ppt. with I_2 and NaOH. Find the number of carbon atoms in principal carbon chain of ester (A).

23. (X) is smallest optically active aldehyde.

$(X) \xrightarrow[\text{conc. HCl}]{Zn-Hg} (Y) \xrightarrow[hv]{Cl_2} \underset{\substack{\text{monochlorinated} \\ \text{products}}}{(Z)} \xrightarrow{\text{fractional distillation}} (N)$

Find the value of (N) if it is the number of fractions obtained on fractional distillation of monochlorinated products (Z).

24. 

How many optically active organic products are formed on free radical monochlorination of optically pure (P) ?

25. $\text{DDT} \xrightarrow[\Delta]{\text{alc. KOH}} \text{Major product}$

Find the number of $sp^2 - p,\ \sigma$- bonds in the major product of the above reaction.

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The area of parallelogram whose diagonals are $\vec{a}$ & $\vec{b}$ is $5\sqrt{3}$. The vectors $\vec{a}, \vec{b}$ can be:

(A) $3\hat{i}+2\hat{j}-\hat{k}, -3\hat{i}-\hat{j}+4\hat{k}$ **(B)** $\frac{3\hat{i}}{2}-\frac{\hat{j}}{2}-\hat{k}, 2\hat{i}-6\hat{j}+8\hat{k}$

(C) $3\hat{i}+\hat{j}-2\hat{k}, \hat{i}+3\hat{j}+4\hat{k}$ **(D)** $-(3\hat{i}+\hat{j}-2\hat{k}), \hat{i}-3\hat{j}-4\hat{k}$

2. Let $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that $|\vec{a}|=3, |\vec{b}|=3, |\vec{c}|=2, |\vec{a}+\vec{b}+\vec{c}|=4$ and $\vec{a}$ is perpendicular to $\vec{b}, \vec{c}$ makes angle θ and ϕ with $\vec{a}$ and $\vec{b}$ respectively, then $\cos\theta+\cos\phi=$

(A) $\frac{3}{4}$ **(B)** $-\frac{3}{4}$ **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

3. The differential equation of the family of curves represented by $y=a+bx+ce^{-x}$ (where, a, b and c are arbitrary constants) is:

(A) $y'''=y'$ **(B)** $y'''+y''=0$ **(C)** $y'''-y''+y'=0$ **(D)** None of these

4. The value of $\int_0^{1.5} x[x^2]\,dx$ (where, [•] denotes the greatest integer function) is:

(A) $\frac{5}{4}$ (B) 0 (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

5. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}-\hat{j}+2\hat{k}$, and $\vec{c}=x\hat{i}+(x-2)\hat{j}-\hat{k}$. If the vector $\vec{c}$ lies in the plane of $\vec{a}$ and $\vec{b}$, then x equals:

(A) 0 **(B)** 1 **(C)** –4 **(D)** –2

6. The area of the region between the curves $y=\sqrt{\frac{1+\sin x}{\cos x}}$ and $y=\sqrt{\frac{1-\sin x}{\cos x}}$ and bounded by the lines $x=0$ and $x=\frac{\pi}{4}$ is:

(A) $\int_0^{\sqrt{2}-1}\frac{t}{(1+t^2)\sqrt{1-t^2}}dt$ **(B)** $\int_0^{\sqrt{2}-1}\frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$

(C) $\int_0^{\sqrt{2}+1}\frac{4t}{(1+t^2)\sqrt{1-t^2}}dt$ **(D)** $\int_0^{\sqrt{2}+1}\frac{t}{(1+t^2)\sqrt{1-t^2}}dt$

7. Let $S_n=\frac{n}{(n+1)(n+2)}+\frac{n}{(n+2)(n+4)}+\frac{n}{(n+3)(n+6)}+......+\frac{1}{(6n)}$, then $\lim_{n\to\infty} S_n$ is equal to:

(A) $\log\left(\frac{3}{2}\right)$ **(B)** $\log\left(\frac{9}{2}\right)$ **(C)** 1 **(D)** 0

8. $\int\left(\frac{\cot^2 2x-1}{2\cot 2x}-\cos 8x\cdot\cot 4x\right)dx$ is equal to:

(A) $\frac{\cos 8x}{8}+C$ **(B)** $\frac{\sin 8x}{8}+C$ **(C)** $-\frac{\cos 8x}{8}+C$ **(D)** $-\frac{\sin 8x}{8}+C$

9. Consider the differential equation $y^2dx+\left(x-\frac{1}{y}\right)dy=0$. If $y(1)=1$, then x is:

(A) $1-\frac{1}{y}+\frac{e^{1/y}}{e}$ **(B)** $4-\frac{2}{y}-\frac{e^{1/y}}{e}$ **(C)** $3-\frac{1}{y}+\frac{e^{1/y}}{e}$ **(D)** $1+\frac{1}{y}-\frac{e^{1/y}}{e}$

10. $\int \frac{x^6(x+1)}{\sqrt{5x^{10}+6x^9+x^4}}dx$ is equal to:

(A) $\frac{\sqrt{5x^6+6x^5+1}}{15}+C$ **(B)** $\frac{\sqrt{5x^6+6x^7+x^2}}{30}+C$

(C) $\frac{\sqrt{5x^8+6x^7+x}}{15}+C$ **(D)** $(5x^{10}+6x^5+x^4)^{1/2}+C$

11. Let $\vec{a}=2\hat{j}-\hat{j}+\hat{k}, \vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=\hat{i}+\hat{j}-2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ *and* $\vec{c}$ whose projection on $\vec{a}$ is $\sqrt{\frac{2}{3}}$ is :

(A) $-(2\hat{i}+3\hat{j}-3\hat{k})$ **(B)** $2\hat{i}-3\hat{j}+6\hat{k}$ **(C)** $-\hat{i}-\hat{j}+2\hat{k}$ **(D)** $3\hat{i}+2\hat{j}-\hat{k}$

12. If $\int (\sqrt{\tan x}+\sqrt{\cot x})dx$ equals:

(A) $\sqrt{2}\sin^{-1}(\cos x-\sin x)+C$ **(B)** $\sqrt{2}\sin^{-1}(\cos x+\sin x)+C$

(C) $\sqrt{2}\tan^{-1}\left(\frac{\sin x+\cos x}{\sqrt{\sin 2x}}\right)+C$ **(D)** $\sqrt{2}\tan^{-1}\left(\frac{\sin x-\cos x}{\sqrt{\sin 2x}}\right)+C$

13. If $(\vec{a}\times\vec{b})\times\vec{c}=\vec{a}\times(\vec{b}\times\vec{c})$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors such that $\vec{a}\cdot\vec{b}\neq 0, \vec{b}\cdot\vec{c}\neq 0$, then $\vec{a}$ and $\vec{c}$ are:

(A) inclined at an angle of $\frac{\pi}{3}$ between them **(B)** inclined at an angle of $\frac{\pi}{6}$ between them

(C) perpendicular **(D)** parallel

14. For $f(x)=x^4+|x|$, let $I_1\int_0^{\pi} f(\cos x)dx$ and $I_2=\int_0^{\pi/2} f(\sin x)dx$, then $\frac{I_1}{I_2}$ has the value:

(A) 1 **(B)** $\frac{1}{2}$ **(C)** 2 **(D)** 4

15. Let $I=\int_0^1\frac{\sin x}{\sqrt{x}}dx$ and $J=\int_0^1\frac{\cos x}{\sqrt{x}}dx$, then which one of the following is true?

(A) $I>\frac{2}{3}$ and $J>2$ **(B)** $I<\frac{2}{3}$ and $J<2$

(C) $I<\frac{2}{3}$ and $J>2$ **(D)** $I>\frac{2}{3}$ and $J<2$

16. Let $f(x)$ be a function defined by $f(x)=\int_1^x t(t^2-3t+2)\,dt, 1\leq x\leq 3$. Then, the range of $f(x)$ is:

(A) $\left[-\frac{1}{4}, 2\right]$ **(B)** $[0, 2]$ **(C)** $\left[-\frac{1}{4}, 0\right]$ **(D)** None of these

17. $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ is equal to:

(A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2 \right\} + C$

(B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7}(\sec x + \tan x)^2 \right\} + C$

(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2 \right\} + C$

(D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7}(\sec x + \tan x)^2 \right\} + C$

18. If $\int_0^1 \frac{e^t}{t+1} dt = a$, then $\int_{b-1}^{b} \frac{e^{-t}}{t-b-1} dt$ is equal to:

(A) ae^{-b} **(B)** $-ae^{-b}$ **(C)** be^{-b} **(D)** None of these

19. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is 30º, then the value of $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is:

(A) $\frac{2}{3}$ **(B)** $\frac{3}{2}$ **(C)** 2 **(D)** 3

20. Let $\vec{a}, \vec{b}$ *and* $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$, the acute angle between $\vec{a}$ and $\vec{c}$ is:

(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{3}$ **(C)** $\frac{\pi}{4}$ **(D)** $\frac{2\pi}{3}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

21. The value of $\frac{5}{\pi}\int_0^{\pi} \frac{x^3 \cdot \cos^4 x \cdot \sin x}{(\pi^2 - 3\pi x + 3x^2)}$

22. If $\vec{a} = \frac{1}{\sqrt{10}}(3\hat{i} + \hat{k})$ and $\vec{b} = \frac{1}{7}(2\hat{i} + 3\hat{j} - 6\hat{k})$, then the value of $(2\vec{a} - \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} + 2\vec{b})]$ is:

23. $\int_{-\pi}^{\pi} \frac{2x \cdot (1 + \sin x)}{1 + \cos^2 x}\, dx = k$, then $[k]$ (where, $[\bullet]$ denotes the greatest integer function) is equal to:

24. If the solution of $\frac{dy}{dx} = (x + y - 1) + \frac{x + y}{\log(x + y)}$ is $1 + \log(x + y) - \log\{k + \log(x + y)\} = x + C$, then k is equal to:

25. Area of the region bounded between the parabola $x^2 = 4y$ and the curve $y = \frac{8}{x^2 + 4}$ is $\left(2\pi - \frac{k}{3}\right)$, then the value of k is:

SPACE FOR ROUGH WORK

End of JEE Main - 13 | JEE 2022